

VibeMonitor — What I Built (9-Month Deep Dive)

 **Role:** Software Engineer (Full-time) · **Period:** September 2025 – June 2026 · **GitHub org:** Vibe-Monitor

VibeMonitor is an AI-driven observability & monitoring platform. My work spans the entire stack — backend, frontend, observability, infrastructure and AI — across the two core repos **vm-api** (Python/FastAPI) and **vm-webapp** (Next.js/React), plus the SDKs, landing site and scheduler.

Contribution at a glance

Metric	Value
Pull requests authored (org-wide)	244 (215 merged)
vm-api (backend)	119 PRs
vm-webapp (frontend)	118 PRs
Other repos	vm-site (4) · scheduler · vm-py-sdk · vm-js-sdk
Active span	12 Sep 2025 → Jun 2026 (~9 months)
Backend modules touched	35+ domain modules

What I built — by theme

1 · CI/CD & DevOps foundation

- Set up the initial CI (ci.yml) and multi-stage ECS deployment workflows; ~20% faster builds via caching, artifact reuse and Docker buildx cache tuning.
- Added an **Alembic migration-validation gate** to the CD pipeline, deployment notifications, change-detection rebuilds, and a GitHub-Actions scheduler for cron jobs.
- Automated RDS snapshot management for backups & environment migrations.
- *Representative PRs:* vm-api#10 — create ci.yml · #93 — scheduler secret token · #234 — force_no_cache + change detection · #350 — Alembic migration validation in CD · #358 — buildx cache mode=min · scheduler#8 — schedule run UI automation

2 · Third-party integrations (GitHub · Slack · Grafana)

- Built the **GitHub App** integration end-to-end: tools router, repository endpoints, webhooks, and GitHub OAuth2 with PKCE.
- Implemented **Slack** (install URL, connect, re-install blocking, auto-uninstall on workspace delete) and **Grafana** integrations.
- *Representative PRs:* vm-api#56 — GitHub App integration · #62 — GitHub tools router · #74 — webhooks · #166 — GitHub OAuth2 + PKCE · webapp#13 — Slack integration · #67 — Grafana integration · #184 — block Slack re-install

3 · Observability provider integrations

- Built credential-verified integrations for **AWS CloudWatch**, **New Relic**, **Datadog**, **Azure Monitor**, **GCP** and **Kubernetes** — logs + metrics APIs for each, with encrypted credential storage and health checks.

- *Representative PRs:* vm-api#106 — AWS · #113 — CloudWatch logs/metrics · #122 — New Relic + encryption · #127/#128 — Datadog · #353 — Azure + Alembic · webapp#140 — Kubernetes connect/disconnect

4 · Agentic RCA engine (AI)

- Wrapped CloudWatch, New Relic, Datadog & Azure as **LangGraph agent tools** so the Root-Cause-Analysis agent can trace incidents through telemetry *and* source code (tree-sitter).
- Fixed agent failure modes (tool-call artifacts, tool-choice=none), and implemented the **HolmesGPT** benchmark — reaching ~90% accuracy on the first 10–20 cases.
- *Representative PRs:* vm-api#138 — CloudWatch + New Relic tools for RCA · #330 — filter tool-call artifacts from RCA output · #341 — 90% accuracy on HolmesGPT · #355 — Azure RCA tools · #337 — RCA benchmark scenario

5 · Authentication & account security

- Reorganised auth into a clean service layer; shipped **Google OAuth**, **GitHub OAuth (PKCE)**, credential-based auth with email verification & password reset, **HttpOnly refresh-token** rotation, "Remember Me", and **Google One-Tap** sign-in.
- *Representative PRs:* vm-api#142 — Google OAuth service · #143 — credential auth · #166 — GitHub OAuth2 PKCE · #297 — HttpOnly refresh tokens · #298 — Remember Me · webapp#39 — Google One-Tap · webapp#28 — email verification + password reset

6 · Billing & monetisation (Stripe)

- Implemented the full **Stripe** billing system: subscription plans, plan up/downgrade logic, service limits, per-key rate limits, and weekly AIU (AI-usage) limits.
- *Representative PRs:* vm-api#222 — whole Stripe functionality · #225 — VibeMonitor rate-limit handling · #239 — service limits Free/Pro · #243 — AIU limits · #245 — downgrade logic

7 · Teams, workspaces, superadmin & enterprise

- Designed the teams & team-membership schema and a full team-management module (CRUD + APIs), plus an enterprise-workspace management service and superadmin tooling.
- *Representative PRs:* vm-api#206 — teams tables & indexes · #209 — team management module · #253 — enterprise workspace service · webapp#88 — enterprise workspaces UI

8 · Blog platform

- Built a complete blog system with author/superadmin permissions and a rich **TipTap** editor — autosave, tables, code-block theming, image upload via an S3 proxy, preview overlay and author roster.
- *Representative PRs:* vm-api#277 — blog system + permissions · #490 — blog image proxy + private S3 · webapp#95 — blog editor & public pages · webapp#177/#178/#179 — autosave, tables, BlockHandle

9 · Log-ingestion platform

- Built high-throughput log ingestion into **ClickHouse** with batching/async pipelines, the LogExplorer UI with bounded time-range queries, plus the dashboard & onboarding flow. Authored the Python & JavaScript SDKs.

- *Representative PRs*: vm-api#420 — logs ingestion · #440 — ClickHouse query optimisation · webapp#148 — Log Ingestion Dashboard & onboarding · webapp#153 — bounded queries · vm-py-sdk#1 · vm-js-sdk#1

10 · Beryl — AI end-to-end test automation (FLAGSHIP)

- Owned much of **Beryl**: crawl a site from one URL → generate natural-language test cases → convert to **Playwright** scripts → run headless → report pass/fail + a site-reliability score.
- Built the streaming onboarding pipeline (crawl → NL-gen → script-gen → run) with **Redis pub/sub**, a parallel worker pool, auto-retry with NL-refactor regeneration, per-step screenshots with visual root-cause, authenticated crawl/auto-login, an SSRF guard and Turnstile captcha.
- *Representative PRs*: vm-api#519 — post-signup tables + signup-claim · #534 — streaming pipeline + Redis pub/sub · #543 — SSRF guard · #555 — auto-retry + NL-refactor · #572 — per-step screenshots + visual RCA · #577 — parallel worker-pool fan-out · #580 — auto-login during crawl · webapp#239 — step-by-step replay view

11 · Frontend platform & UX

- Built large parts of the Next.js app and led the **Redux** → **Zustand** migration; added dark mode, mobile-responsive landing pages (Figma-driven), Error Boundaries, loaders and a silent session-refresh AuthGuard.
- *Representative PRs*: webapp#111 — Redux → Zustand · webapp#176 — dark mode · webapp#122 — mobile responsive · webapp#21 — GlobalError/NotFound · webapp#155 — silent session refresh

12 · Email infrastructure

- Built Mailgun email integration with scheduled sends & templates, then migrated transactional email to **Postmark** templates.
- *Representative PRs*: vm-api#88 — Mailgun + scheduler · #107 — email templates & onboarding logging · #508 — migrate to Postmark



Backend architecture (vm-api module map)

A domain-driven FastAPI codebase — each module is a feature area I worked across.



Month-by-month timeline (from 1:1 journal)

Period	What I shipped
Sep 2025	Onboarding; logs dashboard, signup & workspace-creation flow; learned FastAPI deeply.
Oct 2025	GitHub integration & tools, webhooks, Slack & Grafana; PR-review culture; folder restructure; Mailgun.
Nov 2025	AWS (local + ECS prod), CloudWatch, New Relic; landing pages & beta; Error Boundary; Mailgun templates.
Dec 2025	Datadog; RCA wrappers (CloudWatch/NR/Datadog) + image processing; credential auth; GitHub OAuth; theme redesign; webchat.
Jan 2026	Google One-Tap; LLM config & API-key tracking; team page; preview env via deploy.yml; bug fixes & migrations.

Period	What I shipped
Feb 2026	Full Stripe integration + tests; rate limits; fixed Slack/GitHub bot-uninstall bugs; RED & USE exploration.
Mar 2026	Refresh-token fixes; superadmin/blog; HolmesGPT 90%; K8s; atomic migrations; SOC 2 dependabot; Figma redesign.
Apr–May 2026	Logs-ingestion platform (open-source CloudWatch); Postmark migration; codebase cleanup; Playwright/Beryl scheduler, skip-handling & NL→script; RED/USE blog.
May–Jun 2026	Beryl post-signup flow: streaming pipeline, worker-pool parallelism, auto-retry, screenshots, auto-login crawl, captcha & SSRF guard.

Growth highlights (manager 1:1 feedback)

- Recognised for **consistency, intensity & commitment** ("putting your heart and soul into the work").
- Praised for **integration speed** and being able to add new providers independently.
- Coaching focus I'm acting on: **active listening**, explaining complex reasoning slowly, and writing **structured docs** for clarity.